

Mahmoud Ahmed Elfeky

Software Engineer

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Military Service : Completed



PROFESSIONAL SUMMARY

R&D Software Engineer (Ex-Valeo) with 3 years of experience in C/C++, RTOS, and developing high-reliability systems. This experience is coupled with proven expertise in **complex problem-solving and root-cause analysis**. Possessing a strong technical focus on **IoT Architecture** and **Python Automation** for system efficiency and quality, I am adept at simplifying complex technical concepts for effective troubleshooting and support. Seeking a challenging role in **Embedded Development, IoT, or Technical Instruction**, leveraging deep engineering expertise and passion for knowledge transfer.

WORK EXPERIENCE

- R&D Embedded Software Engineer at Valeo Egypt (Nov 2022 – Oct 2025).
 - Pioneered AI-driven automation tools to optimize R&D workflow and ensure code robustness within the Smart Cockpit/Brain Division.
 - Developed VAAT (Valeo Automated Analysis Tool), an AI-powered C/C++ static analysis agent, integrated with Gemini Code Assist extension, to proactively prevent critical runtime errors.
 - **Scope of Analysis:** Memory Safety (e.g., pointer management), Arithmetic Integrity (e.g., overflows/underflows), Type Cohesion, and Variable Initialization Status.
 - Led the adaptation and integration of an automated, change-based Code Review Tool directly into the build pipeline, resulting in accelerated defect detection and improved overall development efficiency.
 - Software Developer on Driver Monitoring System project with Lucid Motors.
 - Software Developer on Digital Cluster project with Renault.
 - **Integration Test Engineer** on the Digital Cluster project (Stellantis), with Leveraged expertise focusing on internal, external system validation and module integration.
 - **Scope:** Designed, developed, and executed both **manual and automated test cases** (using **Python** and specialized tools) for embedded software systems.
 - **Focus Areas:** Performed **deep integration and regression testing** that covered inter-core communication, inter-module interfaces, and overall **ECU-level system functionality**.
 - **Quality Assurance:** Analyzed test results, meticulously identified software defects, and managed issue tracking to ensure high product quality and compliance.
- Fintech technical support specialist at MaxAB Company in Egypt (Jan 2022- May 2022).

CERTIFICATIONS & DIPLOMAS

- **Full Embedded Systems Diploma covered all these technical skills (165 Hours – Jul 2022) [Certificate:](#)**
 - C Programming and C For Embedded Applications (Embedded C).
 - Micro-controllers Interfacing (Implement all AVR drivers)
 - Full implementation of these peripheral drivers (GPIO, ADC, PWM, ICU)
 - Full implemented of these protocols (SPI, I2C, UART)
 - Real Time Operating Systems (RTOS): Task management, synchronization, priorities, kernel and scheduler operation, mutual exclusion, and deadlock prevention.
 - Software Engineering Fundamentals (SDLC, Waterfall model and V model methodologies).
- **An Introduction to Programming Internet of Things Specialization (144 Hours) | April 2020 [Certificate.](#)**
 - Fundamentals of the Internet of Things (IoT) and Embedded Systems
 - The Raspberry Pi Platform and Python Programming for the Raspberry Pi.
 - Interfacing with the Raspberry Pi.
 - The Arduino Platform and C Programming.
 - Interfacing with the Arduino.
 - Programming for the Internet of Things Final Project.

CORE COMPETENCIES

▪ TECHNICAL SKILLS

- **Programming & Scripting:** C, C++, Embedded C, Python (Automation, Testing, Scripting), Bash/Shell Scripting, MATLAB, Java.
- **Embedded & Automotive:** AUTOSAR, Real-Time Operating Systems (RTOS - Integrity, OSEK), CAN / Ethernet / RTP Protocols, Model-Based Design (MBD).
- **Tools & Environments:**
 - **CI/Automation:** Jenkins, Nexus, Git (Version Control), ITAT.
 - **Analysis & Configuration:** Klockwork, Gemini/GPT, DaVinci (Configurator & Developer), Rhapsody, DOORS, Jira/CORALM.
 - **Testing & Simulation:** CANoe, Proteus Simulator, Lauterbach debugger and TRACE32, Green Hills debugger and Multi IDE.
- **Software Development & Diverse Domains:** Algorithms, Data Structures, Testing, Object-Oriented Programming (OOP), Linux, Android Development, Computer Graphics, Game Development.
- **Specialized Domains:** IoT, Computer Vision, Image Processing, SQL, Compilers.

▪ SOFT SKILLS & INSTRUCTIONAL EXPERTISE

- **Instructional & Leadership:** Highly capable of **teaching, mentoring, and leading technical discussions**. Effective communicator with strong knowledge transfer capabilities.
- **Problem Solving & Analysis:** Proven ability in **root-cause analysis (RCA)**, advanced debugging, and simplifying complex technical processes.
- **Drive & Commitment:** Self-motivated, ambitious, and results-driven with a strong commitment to continuous learning and achieving strategic goals.

EDUCATION

Bachelor's degree in Computer Science

Faculty of Computers and Artificial Intelligence, Benha University, Egypt (2016-2020).

Grade: Very good (84%).

SELECTED PROJECTS :

1. **IoT-Based Smart Agriculture Robot System (Graduation Project) | Grade: Excellent (2020)**

- Designed and implemented a full IoT architecture for automated agriculture monitoring and control.
- Focus: Sensor integration, data analysis, and remote control capabilities.

2. **Python Educational Games Suite (2 Projects)**

- Developed a collaborative Isometric Education Game for teaching programming concepts in Arabic (August 2018).
- Developed a separate 2D Python game focused on children's phonological awareness scale (Jan 2021).
- Technologies: Python, Pygame Module, Computer Graphics.

3. **Embedded Security & Control Systems (4 Projects) | Covered by 2022 Diploma**

- Full implementation of 4 real-time systems (e.g., Password-Based Door Locker, Stop-Watch, Fan Speed Control, Distance Measurement) using Embedded C on ATmega16.
- **Components Used:** Integrated Keypad, EEPROM, Buzzer, DC-Motors, Ultrasonic Sensor, and 7-Segment Displays.
- **Focus Areas:** Complex timing control, external interrupts, and data persistence (EEPROM).

4. **Software Development & Diverse Applications**

- Collaborated with a team of 5 in the development of Digital clock using micro-chips (Dec 2017).
- Collaborated with a team on a GUI University Management System using C++ (Jul 2017).
- Contributed to an Android Development application during summer training (Summer, 2018).
- Implemented a Vehicle Options Control System in C (April 2022).

LEADERSHIP & INSTRUCTIONAL EXPERIENCE

- **Chairman | RETECH | Faculty of Computers and Artificial Intelligence | Benha (2019 – 2020).**
 - Led and managed a technical student organization, overseeing strategic planning, team development, and technical event execution.
- **Python Trainer | Faculty of Computers and Artificial Intelligence | Benha University (2019).**
 - Designed and delivered Python training sessions to university students, demonstrating strong knowledge transfer and instructional capabilities.
- **Head of python team | IEEE BUSB in Shoubra faculty of engineering |Egypt (2018 – 2019).**
 - Mentored and led the Python development team, contributing to educational projects and collaborative technical initiatives.
- **Member of python team | IEEE BUSB in Shoubra faculty of engineering |Egypt (2017 – 2018).**
 - Contributed technical expertise to the Python team's activities and core projects.

LANGUAGES : English (Very Good)